Optimization of algebraic expressions

We already discussed alternative equivalent formulations

$\Pi_{\text{SID,Name}}\ \sigma_{\textit{Teacher}=\text{"}\textit{Meo}\text{"}}$ (student $\bowtie$ exam $\bowtie$ course)

is equivalent to

$\Pi_{\text{SID,Name}}$ (student $\bowtie$ (Π_{SID} (exam $\bowtie$ $\sigma_{\textit{Teacher}=\text{"}\textit{Meo}\text{"}}$ course)))

Equivalent: produces the same result <u>on all possible instances</u>

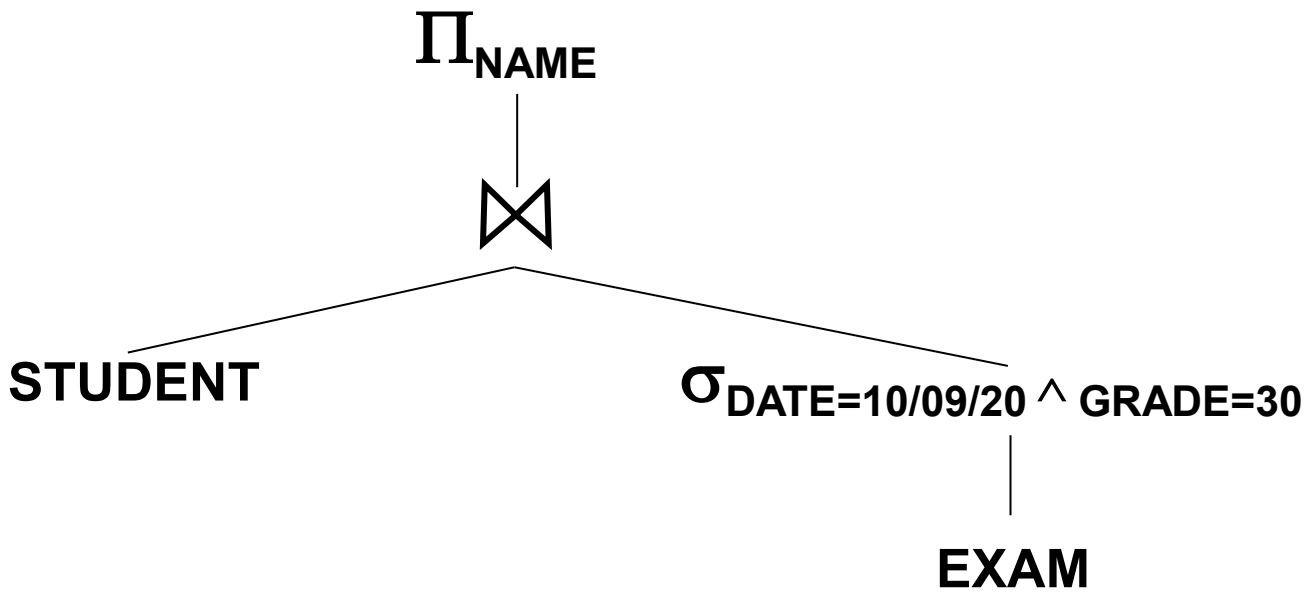
Tree-Syntax tree of algebraic expressions

Any expression can be represented as a tree that expresses the order of application of the operators

Exemple of syntax tree

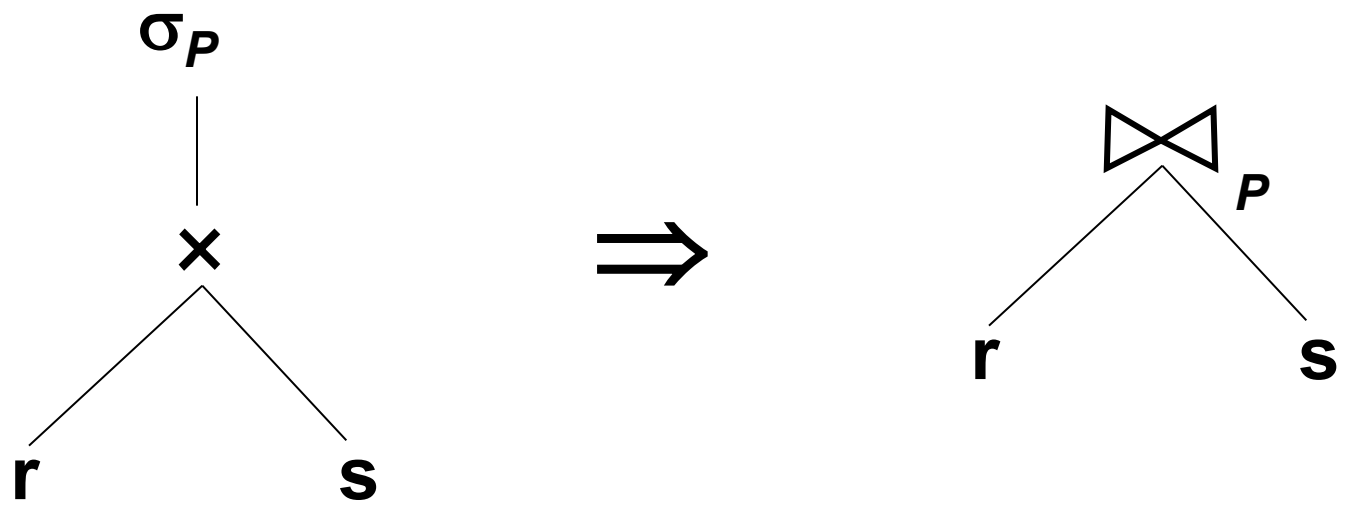
Students who passed an exam on 10/09/20 with a grade equal to 30

$$\Pi_{NAME} (STUDENT \bowtie \sigma_{DATE=10/09/20 \wedge GRADE=30} EXAM)$$



Equivalence of algebraic expressions

1 – ELIMINATION OF CARTESIAN PRODUCTS



True for any P that is a conjunction of comparisons in the form:
ATTRIBUTE <comp> ATTRIBUTE

Equivalence of algebraic expressions

1 – ELIMINATION OF CARTESIAN PRODUCTS

$$\sigma_{student.SID=exam.SID}\ (\ STUDENT\ \times\ EXAM\ )$$

$$\Rightarrow\ STUDENT \bowtie_{student.SID=exam.SID} EXAM$$

MATR	NOME	CITTA'	CL	MATR	C-CORSO	DATA	VOTO
123	Carlo	Bologna	Inf	123	1	09/07/1997	30
123	Carlo	Bologna	Inf	123	2	01/08/1998	28
123	Carlo	Bologna	Inf	702	2	09/07/1997	20
415	Paola	Torino	Inf	123	1	09/07/1997	30
415	Paola	Torino	Inf	123	2	01/08/1998	28
415	Paola	Torino	Inf	702	2	09/07/1997	20
702	Antonio	Roma	Log	123	1	09/07/1997	30
702	Antonio	Roma	Log	123	2	01/08/1998	28
702	Antonio	Roma	Log	702	2	09/07/1997	20

STUDENT

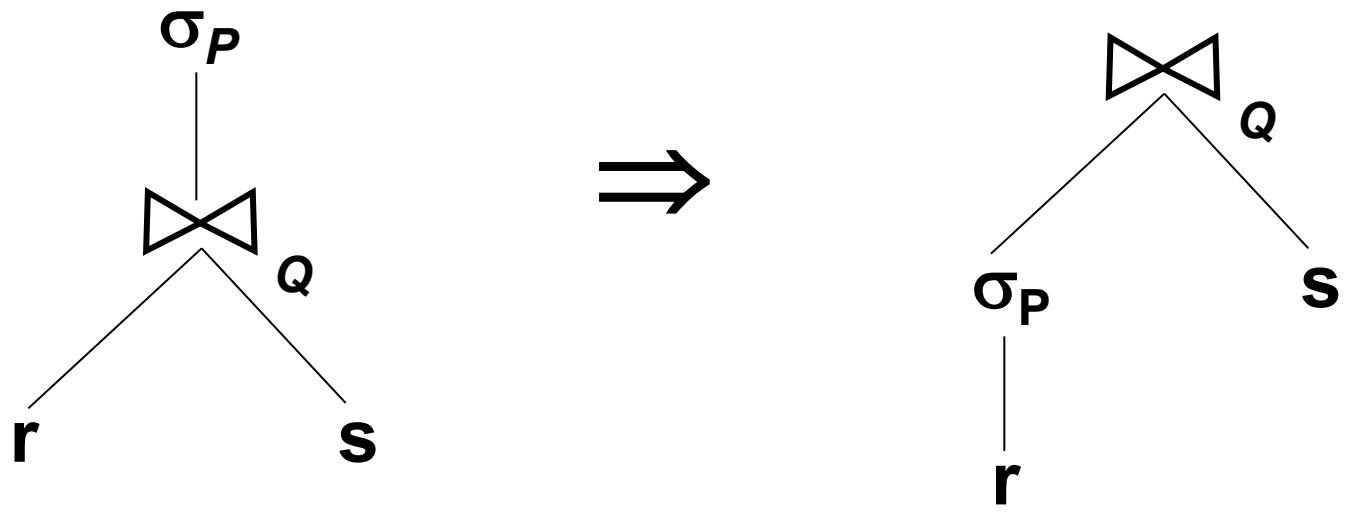
SID	NAME	CITY	DEP
123	Carlo	Bologna	Inf
415	Paola	Torino	Inf
702	Antonio	Roma	Log

EXAM

SID	CID	DATE	GRADE
123	1	7/9/1997	30
123	2	8/1/1998	28
702	2	7/9/1997	20

Equivalence of algebraic expressions

2 - PUSH OF SELECTION THROUGH JOINS



True for any P that only targets attributes of r.
Or we can split the conjuncts on the two branches.

Equivalence of algebraic expressions : example

2 - PUSH OF SELECTION THROUGH JOINS

$$\sigma_{City=«Torino»}\ (\ STUDENT \bowtie EXAM\ )$$

$$\Rightarrow\ (\sigma_{City=«Torino»}\ STUDENT\ ) \bowtie EXAM$$

STUDENT

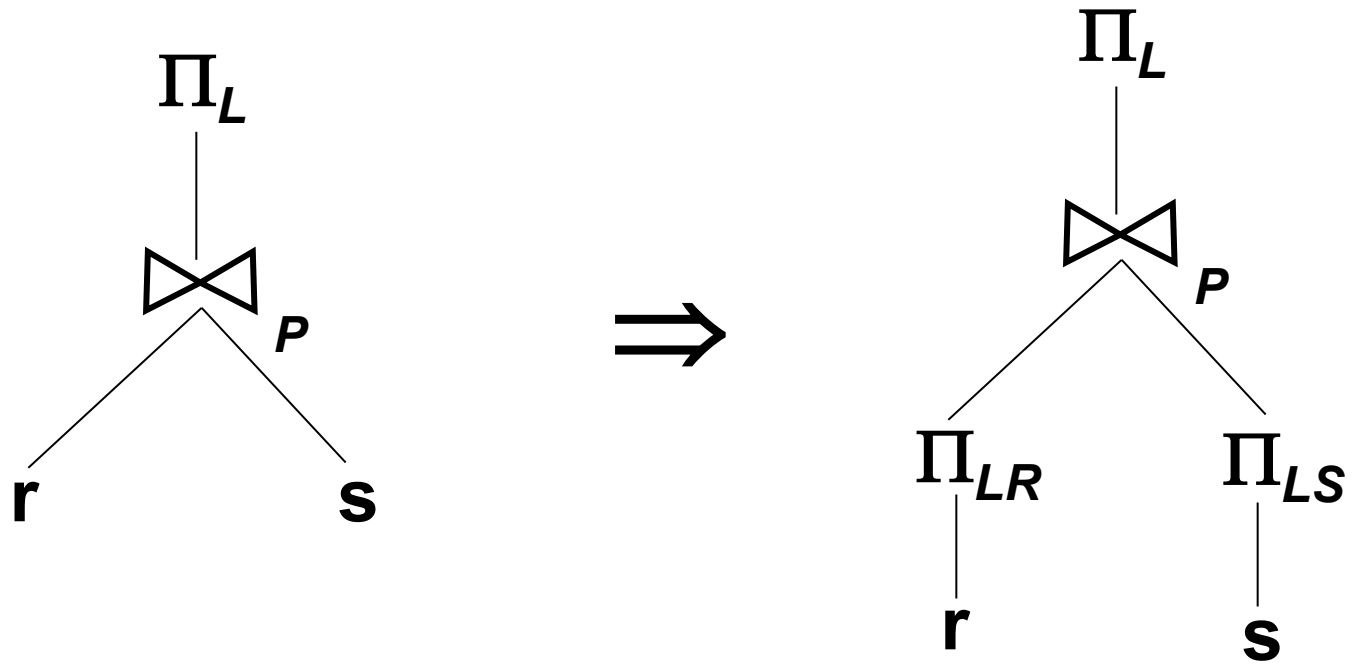
SID	NAME	CITY	DEP
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415	Paola	Torino	Inf
702	Antonio	Roma	Log

EXAM

SID	CID	DATE	GRADE
123	1	7/9/1997	30
123	2	8/1/1998	28
702	2	7/9/1997	20

Equivalence of algebraic expressions

3 - PUSH OF PROJECTION THROUGH JOINS



Where

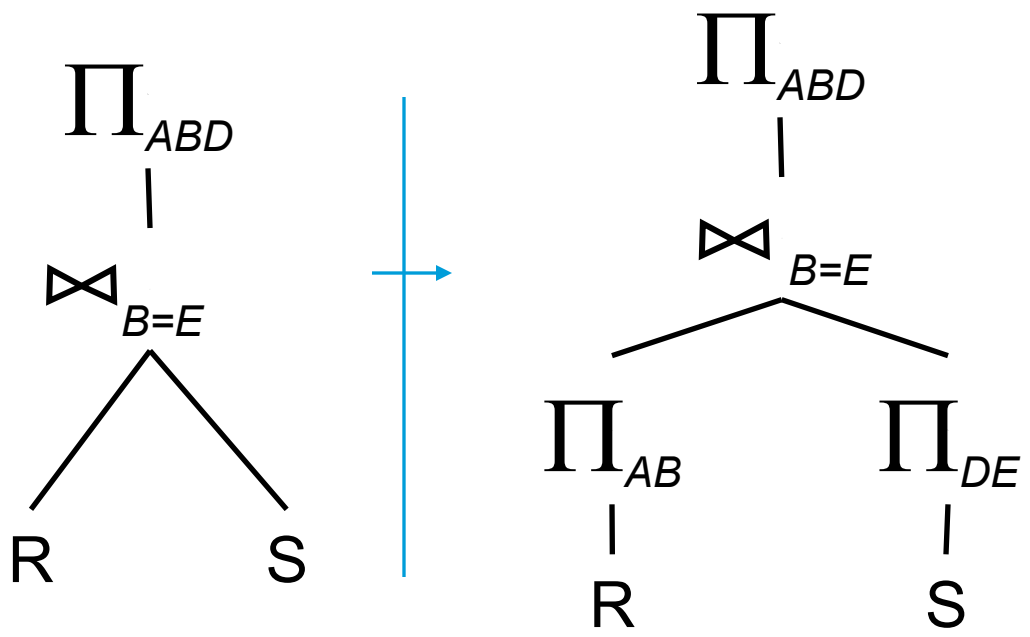
***LR* = *L* - SCHEMA(s) + JR**

***LS* = *L* - SCHEMA(r) + JS**

with JR and JS being the attributes of R and S required in P

Equivalence of algebraic expressions : example

3 - PUSH OF PROJECTION THROUGH JOINS



R(A,B,C)
S(D,E,F,G)

L = ABD

JR = B
JS = E

LR = AB
LS = DE

Equivalence of algebraic expressions : example

3 - PUSH OF PROJECTION THROUGH JOINS

$$\Pi_{SID,Name}\ (\ STUDENT\ \bowtie\ EXAM\ )$$

$$\Rightarrow\ (\ \Pi_{SID,Name}\ STUDENT\ )\ \bowtie\ (\ \Pi_{SID}\ EXAM\ )$$

STUDENT

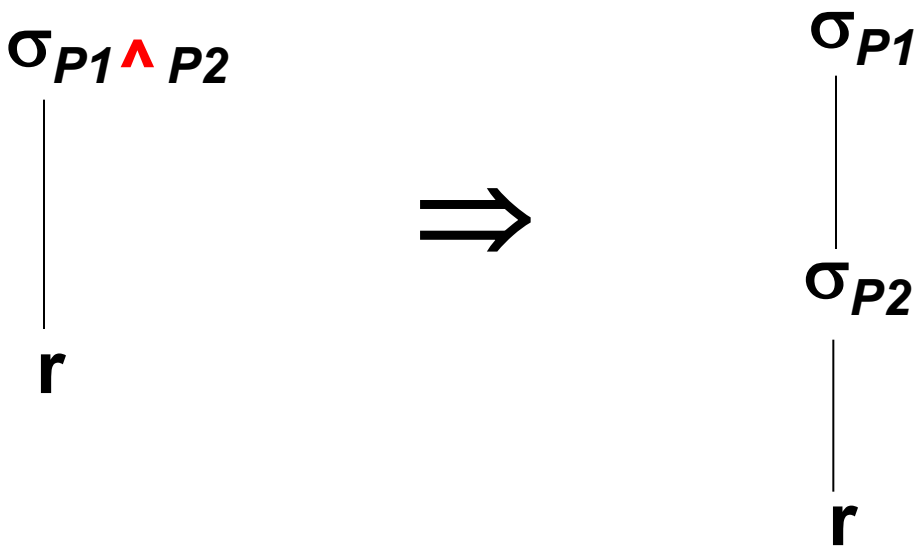
SID	NAME	CITY	DEP
123	Carlo	Bologna	Inf
415	Paola	Torino	Inf
702	Antonio	Roma	Log

EXAM

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123	1	7/9/1997	30
123	2	8/1/1998	28
702	2	7/9/1997	20

Equivalence of algebraic expressions

4 - IDEMPOTENCE OF SELECTION



Equivalence of algebraic expressions : example

Combining idempotence of selection (4)

and push of selection through joins (2)

$$\sigma_{City=«Torino» \wedge Grade=«30»} (STUDENT \bowtie EXAM)$$

$$\Rightarrow \sigma_{City=«Torino»} (\sigma_{Grade=«30»} (STUDENT \bowtie EXAM))$$

$$\Rightarrow (\sigma_{City=«Torino»} STUDENT) \bowtie (\sigma_{Grade=«30»} EXAM)$$

STUDENT

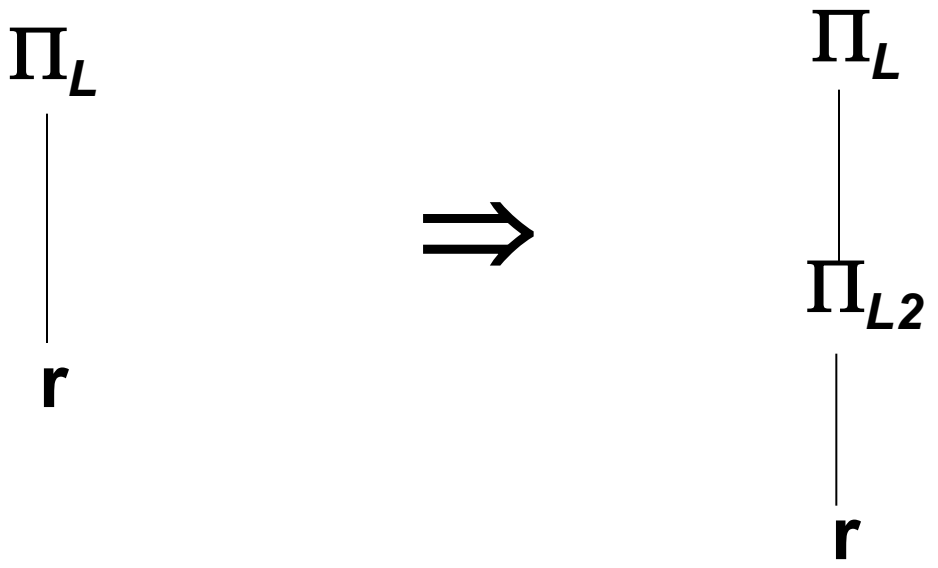
SID	NAME	CITY	DEP
123	Carlo	Bologna	Inf
415	Paola	Torino	Inf
702	Antonio	Roma	Log

EXAM

SID	CID	DATE	GRADE
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123	2	8/1/1998	28
702	2	7/9/1997	20

Equivalence of algebraic expressions

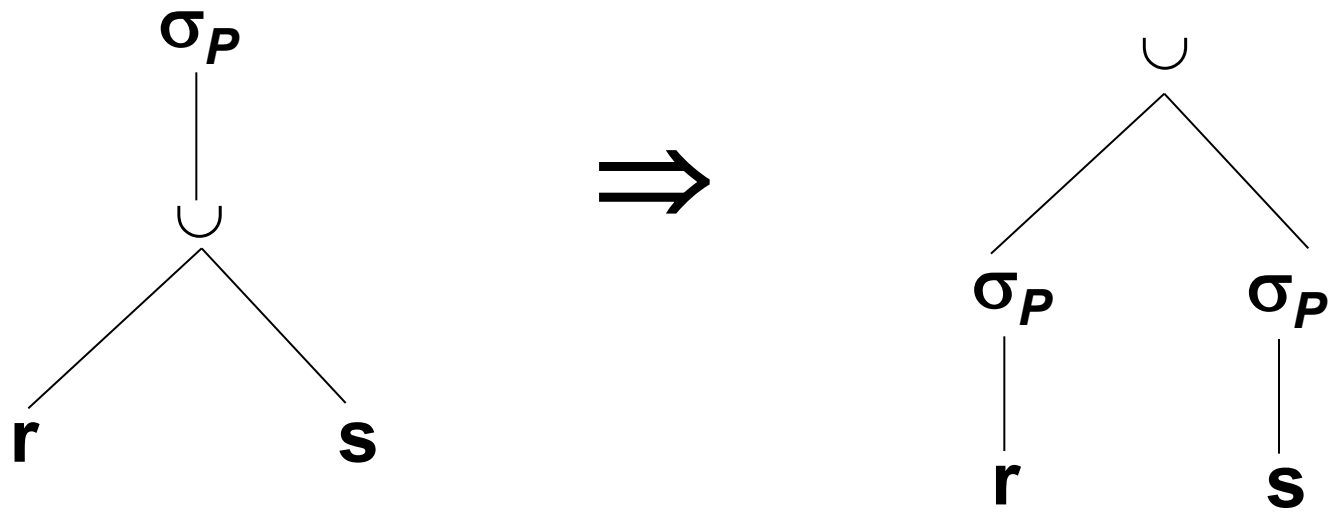
5 - IDEMPOTENCE OF PROJECTION



with L2 ⊇ L

Equivalence of algebraic expressions

6 - PUSH OF SELECTION THROUGH UNION



Equivalence of algebraic expressions : example

6 - PUSH OF SELECTION THROUGH UNION

$$\sigma_{Name=«Paola»}\ (\ ENGINEERS\ \cup\ TURINERS\ )$$

$$\Rightarrow\ (\sigma_{Name=«Paola»}\ ENGINEERS\ )\ \cup\ (\sigma_{Name=«Paola»}\ TURINERS\ )$$

STUDENT

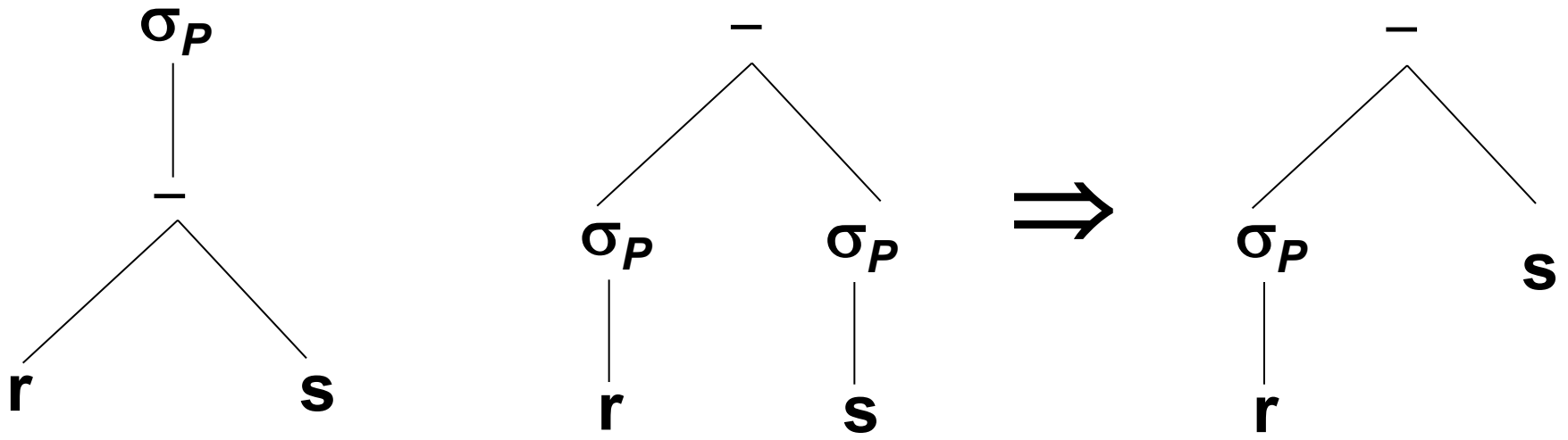
SID	NAME	CITY	DEP
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415	Paola	Torino	Inf
702	Antonio	Roma	Log

EXAM

SID	CID	DATE	GRADE
123	1	7/9/1997	30
123	2	8/1/1998	28
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Equivalence of algebraic expressions

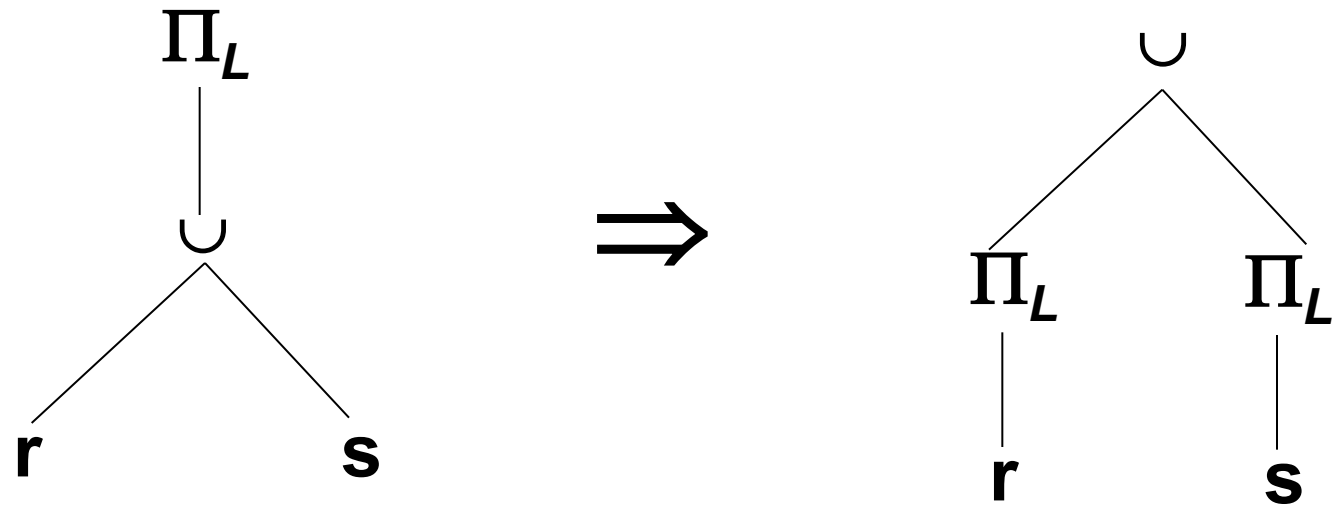
7 - PUSH OF SELECTION THROUGH DIFFERENCE



Selection commutes with all three set operators (∩ as well)

Equivalence of algebraic expressions

8 - PUSH OF PROJECTION THROUGH UNION



N.B.: projection does NON commute with the other two set operators (– and ∩)

Equivalence of algebraic expressions

Projection does NON commute with the other two set operators (– and ∩)

Difference: $R_1(A,B)$, $R_2(A,B)$

$\Pi_A(R_1 - R_2)$ is NOT equivalent to $\Pi_A R_1 - \Pi_A R_2$

(see next slide)

Intersection: obviously does not commute (as it is a derived operator, from ∪ and –)

Projection does not commute with difference

Clients_of_branch_A

CLIENT	EMPLOYEE
Rossi	Ghezzi
Brambilla	Ferrari
Verdi	Ferrari

Clients_of_branch_B

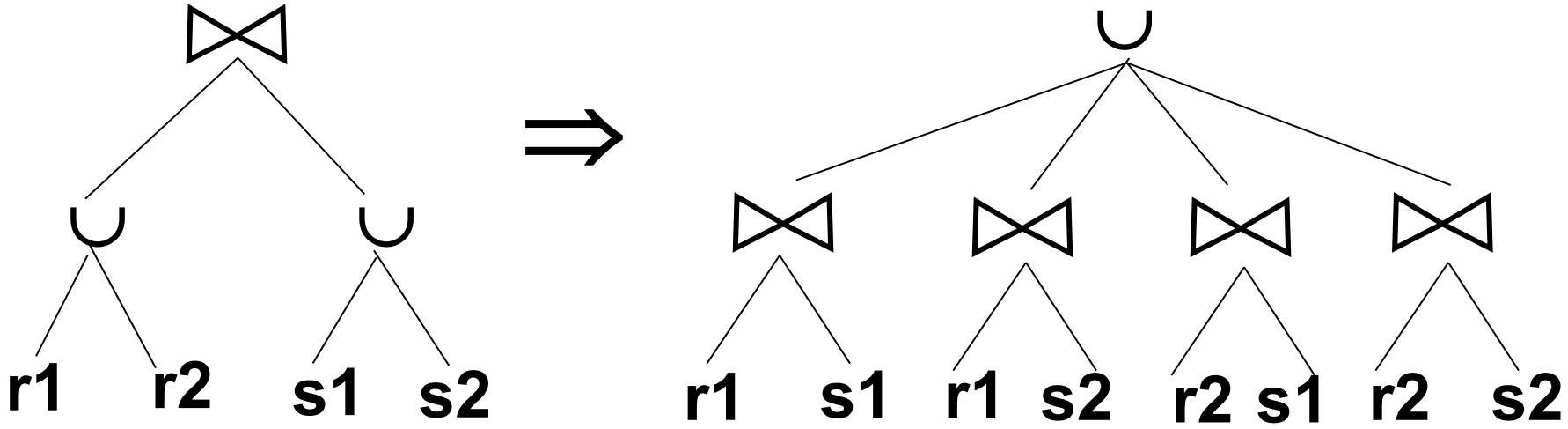
CLIENT	EMPLOYEE
Rossi	Amato
Brambilla	Ferrari
Verdi	Lupo

$$\Pi_{\text{CLIENT}} (\text{ClientsA} - \text{ClientsB}) = \{ \textbf{Rossi, Verdi} \}$$

$$\Pi_{\text{CLIENT}} \text{ClientsA} - \Pi_{\text{CLIENT}} \text{ClientsB} = \varnothing$$

Equivalence of algebraic expressions

9 – Distribution of JOIN over UNION



Useful formulas

$r \bowtie r = r$

$r \cup r = r$

$r - r = \varnothing$

$r \bowtie \sigma_P r = \sigma_P r$

$r \cup \sigma_P r = r$

$r - \sigma_P r = \sigma_{\neg P} r$

$\sigma_{P1} r \bowtie \sigma_{P2} r = \sigma_{P1 \wedge P2} r$

$\sigma_{P1} r \cup \sigma_{P2} r = \sigma_{P1 \vee P2} r$

$\sigma_{P1} r - \sigma_{P2} r = \sigma_{P1 \wedge \neg P2} r$

$\sigma_P \varnothing = \varnothing$

$\Pi_L \varnothing = \varnothing$

$r \times \varnothing = \varnothing$

$r \cup \varnothing = r$

$r - \varnothing = r$

$\varnothing - r = \varnothing$

$r \bowtie \varnothing = \varnothing$

$r \cap \varnothing = \varnothing$

As you can see, the «same» formulas are valid for join and intersection

Algebraic Optimization

Of all equivalent representations, we shall favor the one with the **minimum execution cost**. But how can we assess the execution cost?

- Informal criterion: minimize the **size** of intermediate results
  - Apply the maximum amount of push (2,3,6,7,8)
  - Use idempotence (4, 5) to generate new selections and projections
- Combine as much as possible Cartesian products and selections any time they can become joins

Example of Optimization

SCHEMA

R(A,B,C)

S(C,D,E)

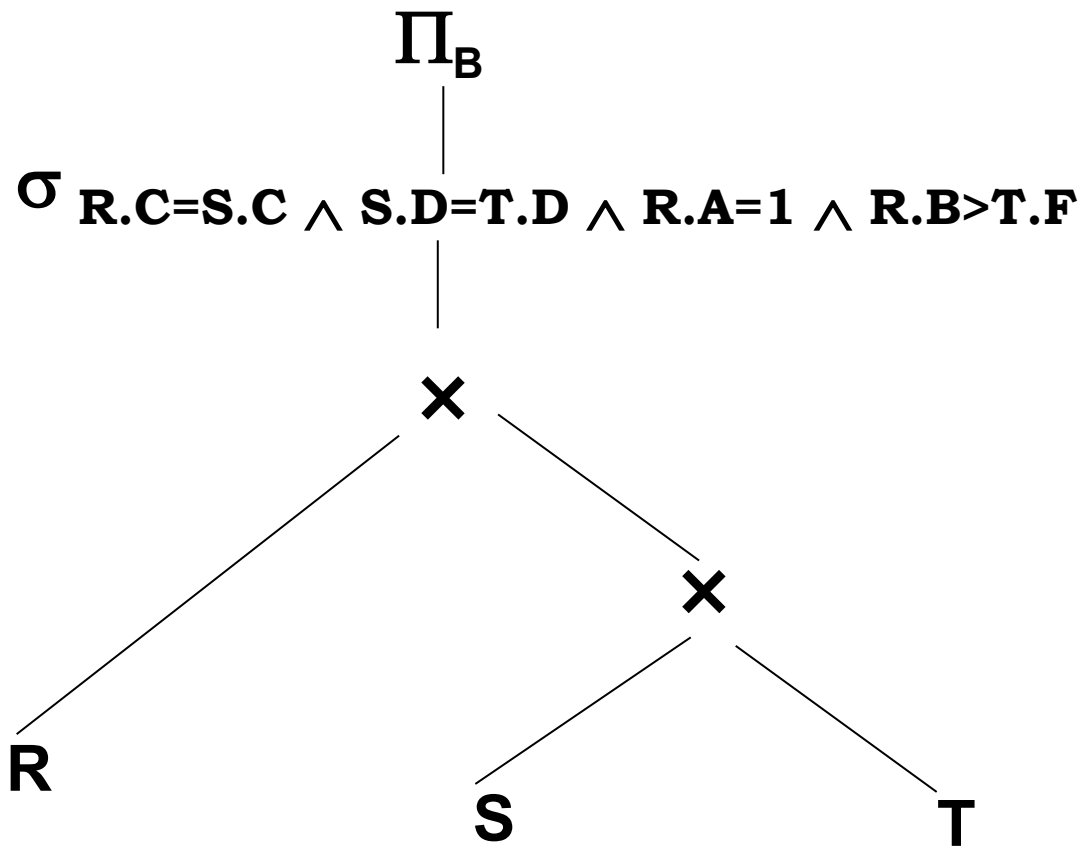
T(D,E,F,G)

Optimize the following expression:

$$\Pi_{B}\,\sigma_{(R.C=S.C)\,\wedge\,(S.D=T.D)\,\wedge\,(R.A=1)\,\wedge\,(R.B>T.F)}\,(R \times S \times T)$$

Example of Optimization

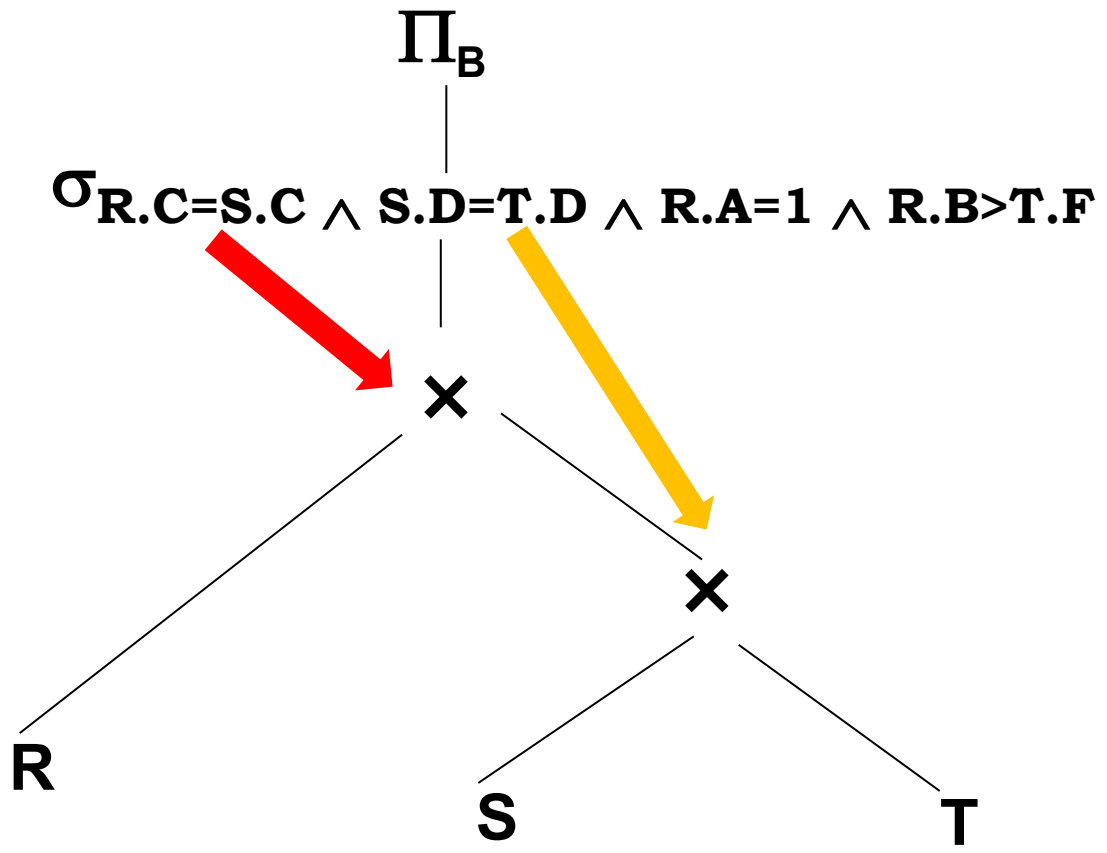
$$\Pi_B \, \sigma_{(R.C=S.C) \wedge (S.D=T.D) \wedge (R.A=1) \wedge (R.B>T.F)} \, (R \times S \times T)$$



R (A,B,C)
S (C,D,E)
T (D,E,F,G)

Example of Optimization

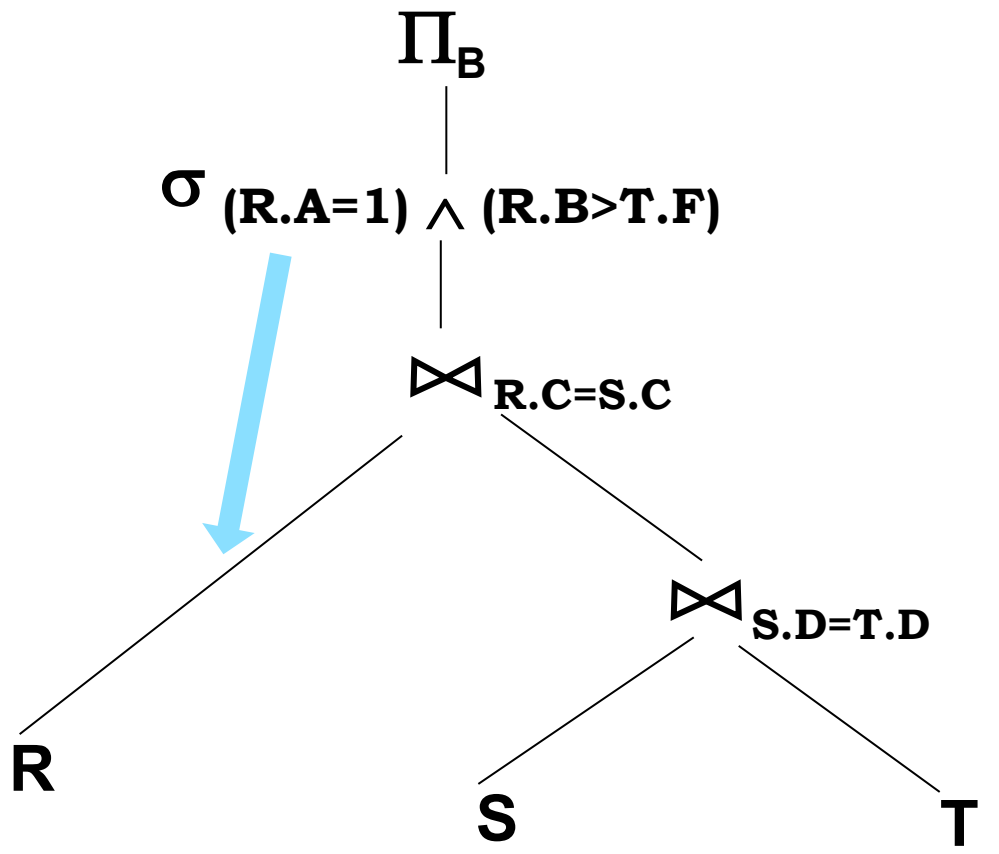
$$\Pi_B \, \sigma_{(R.C=S.C) \wedge (S.D=T.D) \wedge (R.A=1) \wedge (R.B>T.F)} \, (R \times S \times T)$$



R (A,B,C)
S (C,D,E)
T (D,E,F,G)

Example of Optimization

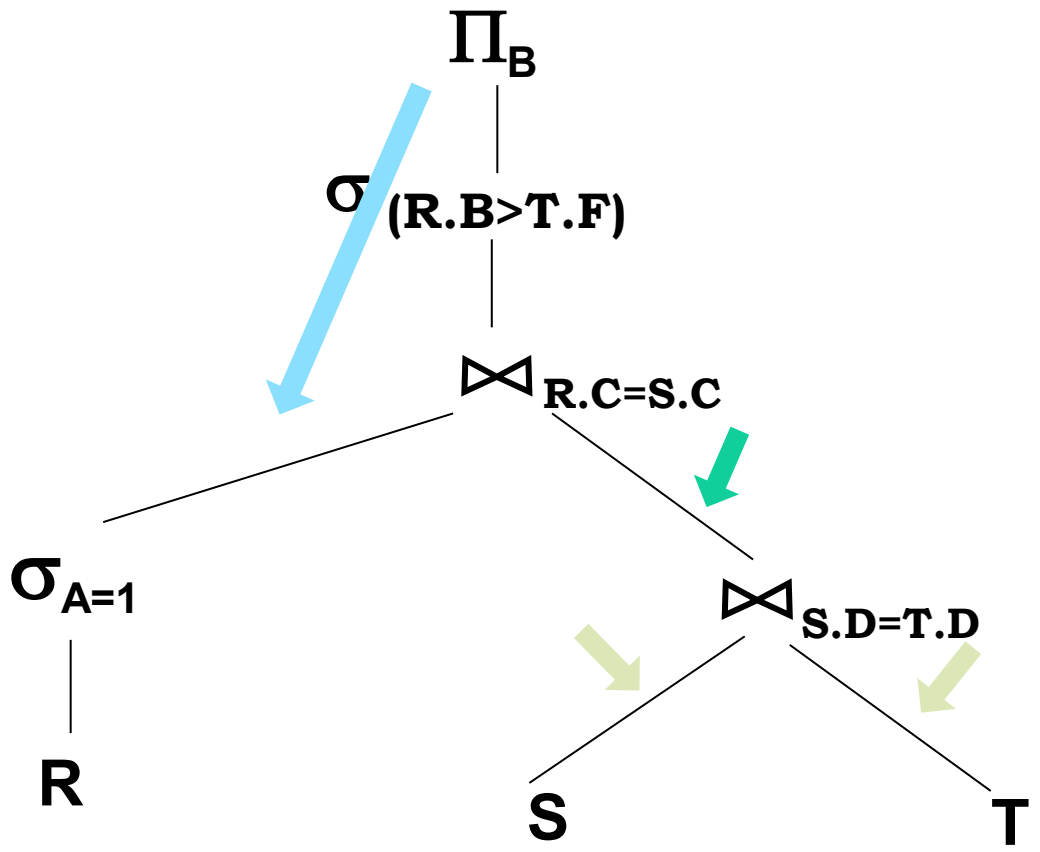
$$\Pi_B \, \sigma_{(R.C=S.C) \wedge (S.D=T.D) \wedge (R.A=1) \wedge (R.B>T.F)} \, (R \times S \times T)$$



R (A,B,C)
S (C,D,E)
T (D,E,F,G)

Example of Optimization

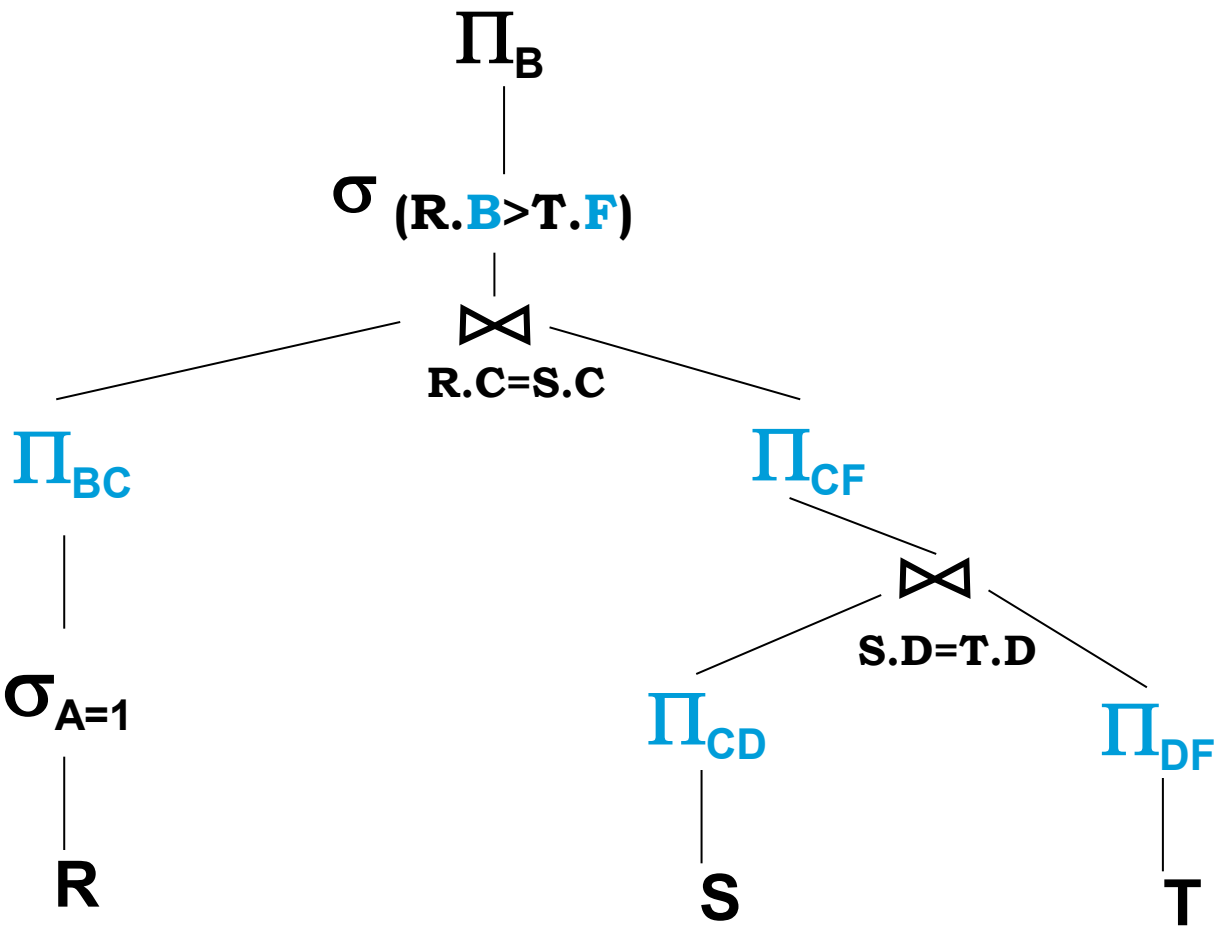
$$\Pi_B \, \sigma_{(R.C=S.C) \wedge (S.D=T.D) \wedge (R.A=1) \wedge (R.B>T.F)} \, (R \times S \times T)$$



R (A,B,C)
S (C,D,E)
T (D,E,F,G)

Example of Optimization

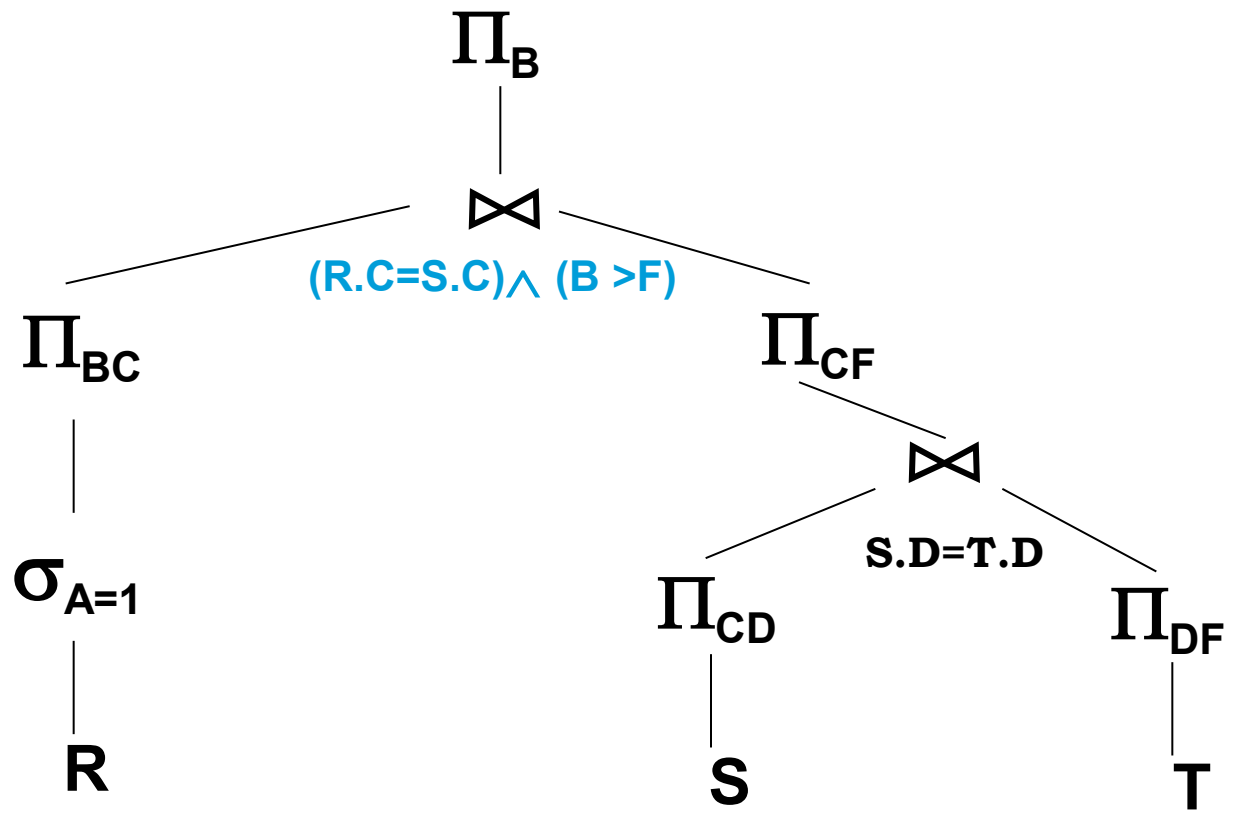
$$\Pi_B\, \sigma_{(R.C=S.C) \wedge (S.D=T.D) \wedge (R.A=1) \wedge (R.B>T.F)} \; (R \times S \times T)$$



R (A,B,C)
S (C,D,E)
T (D,E,F,G)

Example of Optimization

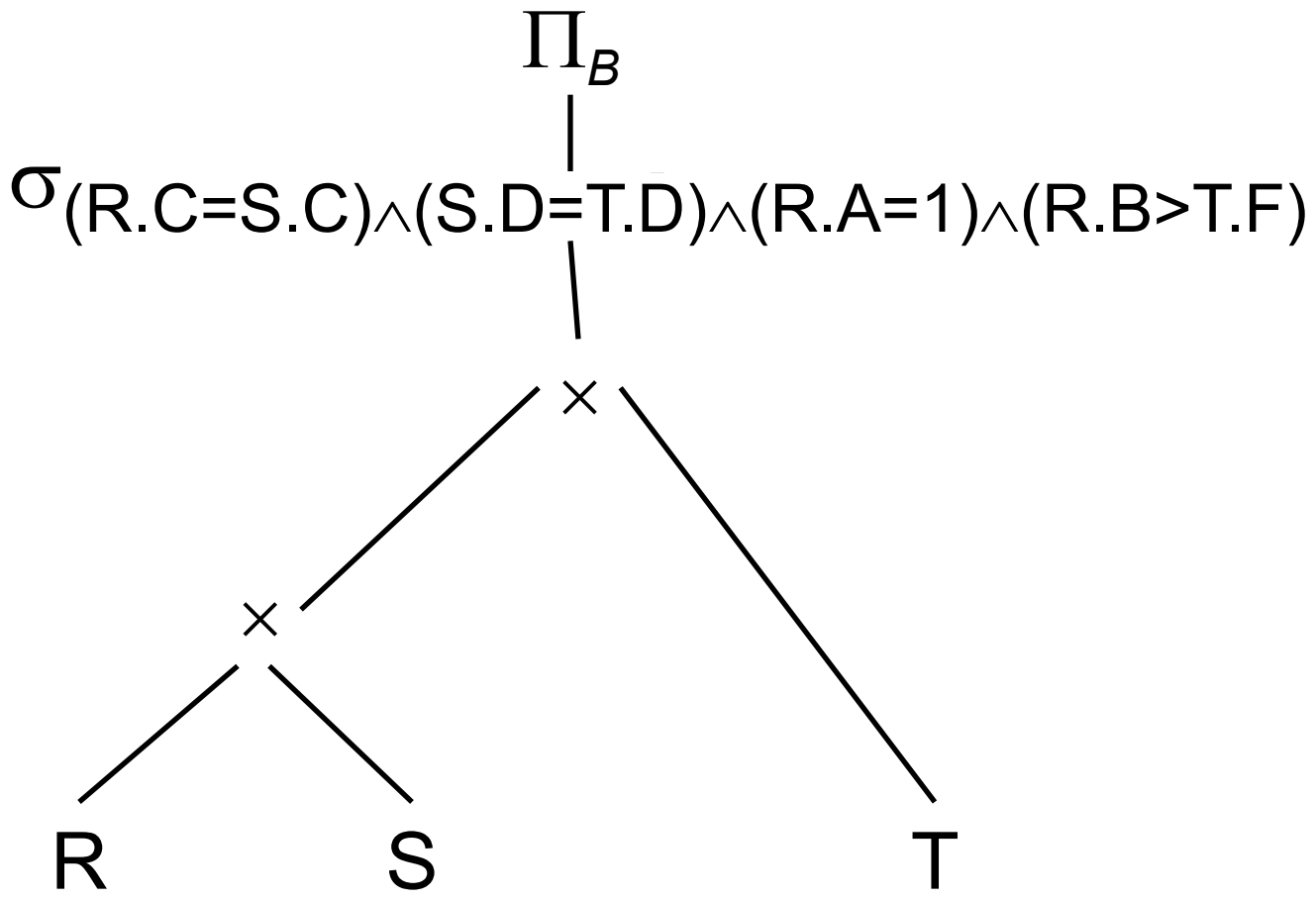
$$\Pi_B\, \sigma_{(R.C=S.C) \wedge (S.D=T.D) \wedge (R.A=1) \wedge (R.B>T.F)} (R \times S \times T)$$



R (A,B,C)
S (C,D,E)
T (D,E,F,G)

Exercise

- Repeat with:



Optimization Exercises

1) Given the schema: R(A,B)
S(A,B)

optimize:

$\sigma_{(S.A=R.A)\wedge(R.A>2)\wedge(S.A=1)}\ R \times S$

….shall we realize at once that the result is $\varnothing$??

Why?

Optimization Exercises

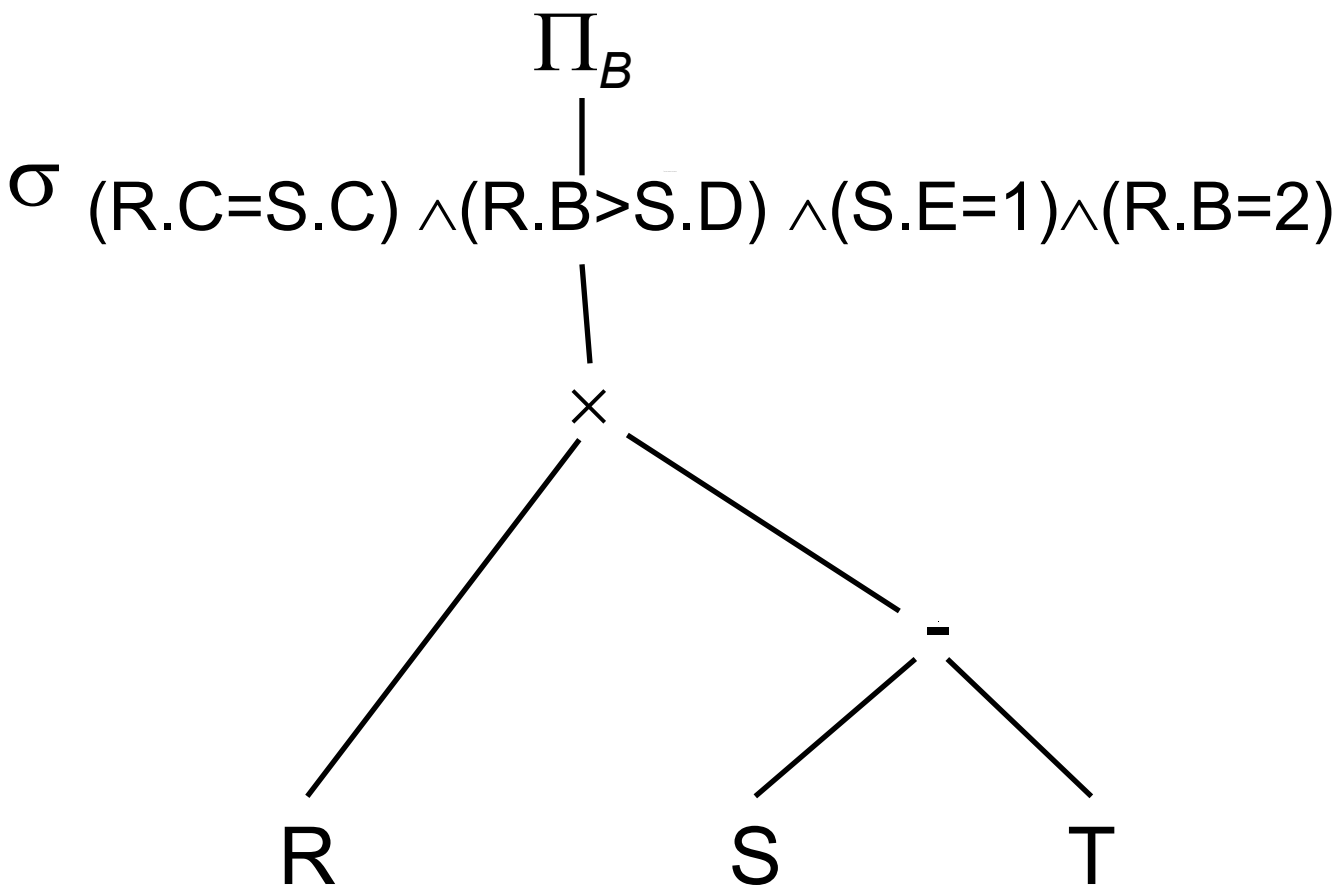
2) Given the schema: R(A,B,C)
S(C,D,E)
T(C,D,E)

Optimize:

$$\Pi_B \, \sigma_{(R.C=S.C)\wedge(S.E=1)\wedge(R.B=2)\wedge(R.B>S.D)} \, (R \times (S - T))$$

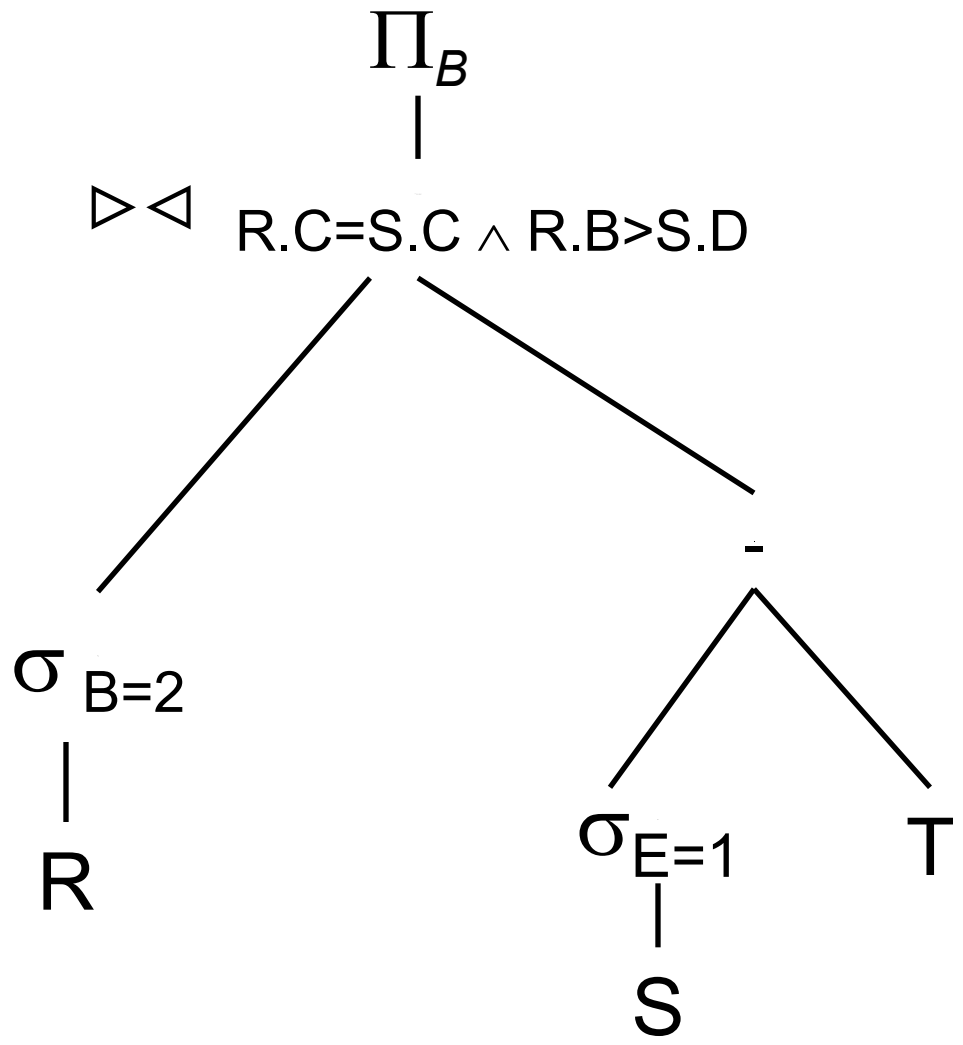
Example of Optimization

- Represented as a syntax tree:



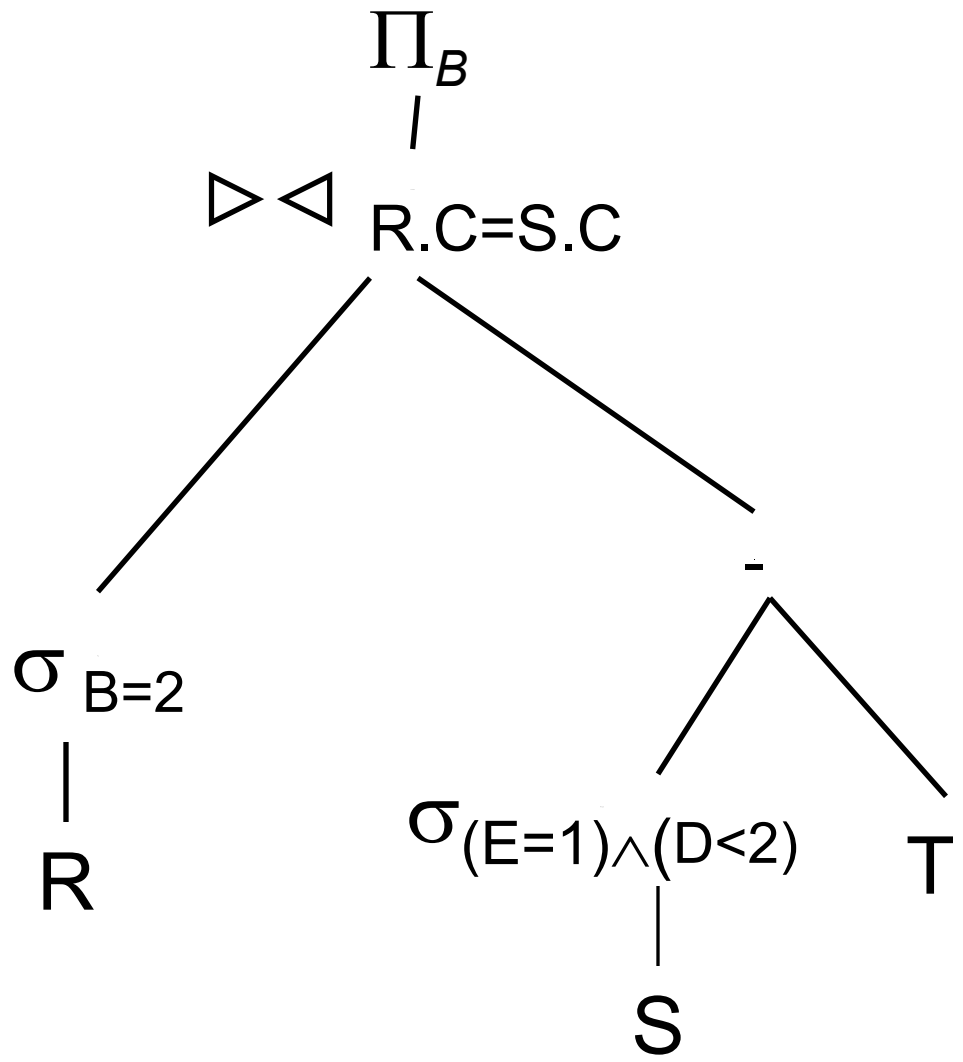
Example of Optimization

- With the «automatic» transformations we get:



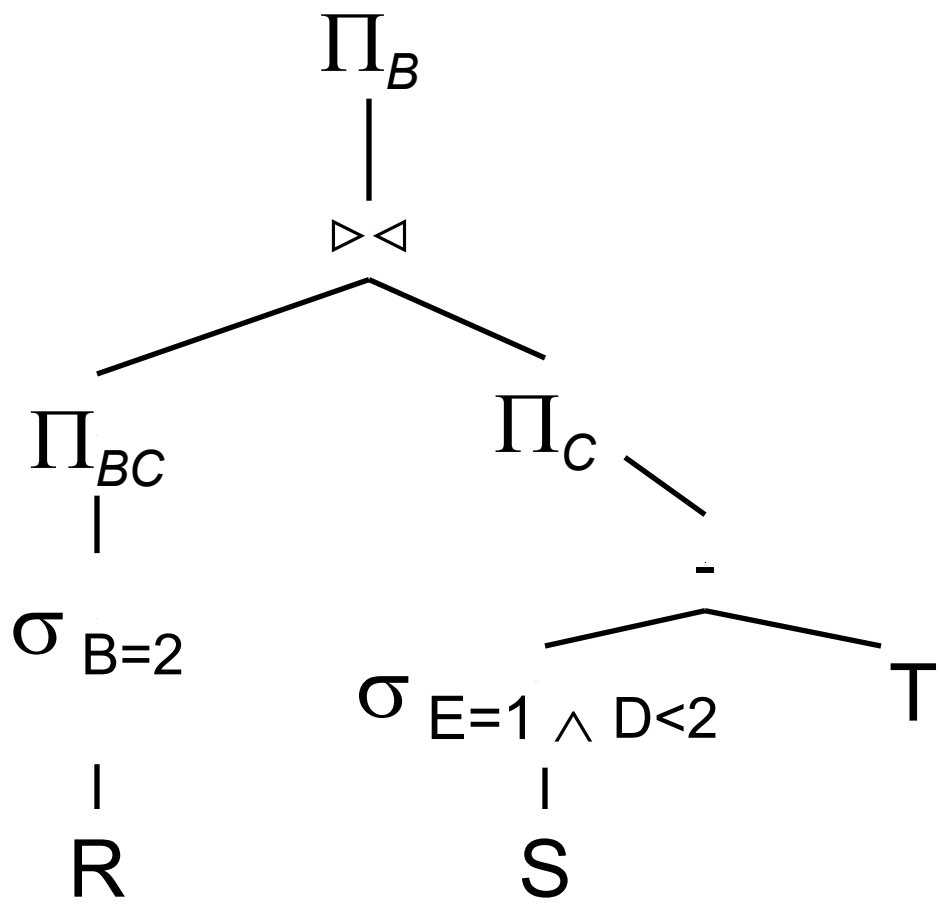
Example of Optimization

- Reasoning on the predicates we can furtherly simplify:



Example of Optimization

- Adding the intermediate projections, we finally get:



RelaX - relational algebra calculator

calculates any relational algebra statement like $(\sigma_{a > 42} (A)) \bowtie (\pi_{a, b} (B))$ on a set of relations.

http://dbis-uibk.github.io/relax/

